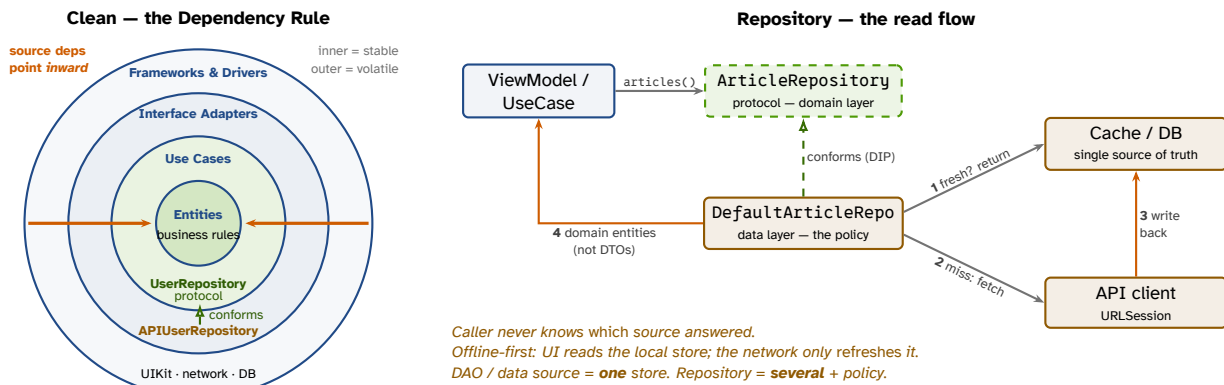


In one line: Four ways to stop one object knowing too much. **Coordinator** — *where to go next* (navigation out of VCs). **Repository** — *where data comes from* (hide network/cache/DB behind one protocol). **DI** — *who builds my collaborators* (someone outside, at the composition root). **Clean** — *which way dependencies point* (always inward, towards business rules).



How it works

Coordinator. *Problem:* VCs that push/present each other are coupled to their siblings, cannot be reused in another flow, bloat with routing, and make deep links scatter. **3 jobs:** (1) **build** screens — VC+VM, inject deps (assembly); (2) **navigate** — push/present/dismiss on its presenter (UINavigationController); (3) **own child flows** — childCoordinators, start(), remove on finish (forget it → the flow leaks). The VC only reports intent (closure onSelect, delegate) — it is a *Mediator* for navigation. SwiftUI: a router owning NavigationPath/[Route] + navigationDestination(for:).

Repository. One source-agnostic API over *several* sources (network, cache, DB); orchestrates them with a policy (cache → API → write-back, merge, staleness); returns **domain entities**; is swappable behind a protocol the *domain* owns. Caching/offline lives *here*.

Dependency Injection = collaborators handed in from outside.

- **Constructor** (init) — default: non-optional, let, compiler-enforced. **Property** — optional deps, storyboard VCs (a nil window). **Method** — per-call dependency.
- **Default arg:** init(s: URLSession = .shared) — tests inject.
- **Containers:** Swinject, Resolver, Factory, Needle (compile-time); scopes: transient · singleton · scoped/weak.
- **SwiftUI:** .environment(\.key, v) / .environmentObject(o) — DI by context; missing → runtime crash.
- **Composition root:** the one place near launch that wires concretes. **Service locator** (a container resolved *inside* a type) = anti-pattern: hidden deps, global state.
- **IoC** = idea · **DIP** = principle · **DI** = technique implementing DIP.

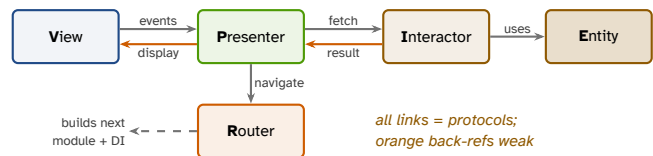
Remember

Coordinator = *where to* · **Repository** = *where from* · **DI** = *who builds* · **Clean** = *which way*. VIPER = V-I-P-E-R.

Example

```
protocol ArticleRepository {           // owned by the DOMAIN
    func articles() async throws -> [Article]
}
final class DefaultArticleRepo: ArticleRepository {
    let api: APIClient; let cache: ArticleCache
    init(api: APIClient, cache: ArticleCache) { // ctor DI
        self.api = api; self.cache = cache
    }
    func articles() async throws -> [Article] {
        if let c = cache.load(), !c.isStale { return c.items }
        let fresh = try await api.fetch().map(Article.init)
        cache.save(fresh); return fresh // write back
    }
}
// composition root: the ONLY place that picks concretes
FeedVM(repo: DefaultArticleRepo(api: LiveAPI(), cache: Disk()))
```

VIPER, one module per screen



I = business logic (use case) · **P** = formatting · **R** = navigation + assembly · **E** = plain models. **I** ≈ Clean's Use Case, **P** ≈ Interface Adapter. objc.io's original: I hands P **plain data, never Entities** (so P can only format); the wireframe (R) owns the window / UINavigationController. Ownership: V→P→I and P→R strong; P→V, I→P, R→VC **weak** (protocols : AnyObject) or the module never frees. Cost: ~5 types + protocols *per screen* — only for large, multi-team apps.

Traps

- ! Repository is **not mainly about testing**: it hides *multiple* sources · read flow **cache** → **API** → **write-back** · **single source of truth**. Testability falls out.
- ! A class that just forwards api.get() is a renamed network client, not a Repository — the value is orchestration.
- ! Coordinator as “a class that pushes VCs” misses the point — the **problem** is VCs knowing each other (coupling, no reuse); the 3 jobs follow.
- ! .shared called inside a type defeats DI — inject it.
- ! “Use case needs network so it imports URLSession” — no: it declares a protocol; the outer layer conforms and is injected.